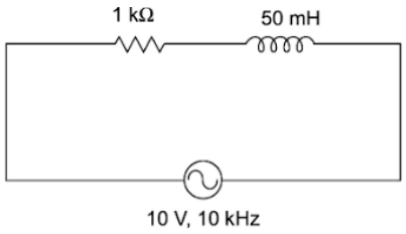
DHOLE PATIL EDUCATION SOCIETY'S
DHOLE PATIL COLLEGE OF ENGINEERING

Department of First year Engineering

Course Code: ESC-102-ELE | **Subject: Basic Electrical Engineering** | **Academic Year – 2025-26**

Question Bank | **Semester -I**

Unit –IV AC CIRCUITS	
S.NO:	**QUESTIONS**
Que.1	Derive the expression for resonant frequency and comment on impedance, current and power factor in case of series resonance circuit
Que.2	A delta connected balanced load consists of three coils; each of resistance 6 Ω and inductive reactance 8 Ω supplied by 400V, 3-phase AC. Determine i) impedance per phase ii) phase current and line current iii) power factor of the circuit iv) total power absorbed
Que.3	Obtain the expressions for instantaneous power and average power in a series R-L circuit excited by purely sinusoidal voltage
Que.4	An R-C series circuit dissipates 100 W at 0.75 leading power factor when connected across single phase. 230 V. 50 Hz sinusoidal AC supply. Calculate: i) current drawn ii) impedance in polar and rectangular form iii) resistance of the circuit and iv) value of capacitance
Que.5	Derive the expression for power, when voltage v = Vm sin ωt is applied across R-L series circuit. Also draw the waveform for voltage, current & power on common X-axis
Que.6	The series circuit having resistance 5 Ω, inductance 0.1 H and capacitance 150 μF is connected to 1-phase, 200 V, 50 Hz AC supply. Calculate- i) Inductive reactance XL ii) Capacitive reactance Xc iii) Net reactance X iv) Impedance Z v) Current drawn by the circuit vi) Power factor vii) Active power P viii) Reactive power Q
Que.7	Draw phasor diagram for purely inductive and capacitive circuit. There are two impedances Z1= 3 +j 4 and Z2 = 6 – j 8. Carry out following mathematical operations to find Z (equivalent) in rectangular form. 1) Z1 × Z2 2) Z2 – 3 Z1 3) Z2 – 2 Z1
Que.8	Derive the relation between line voltage and phase voltage for three phase star connected balanced R-L load.
Que.9	**Explain power triangle in R-L circuit and estimate three types of powers in a typical motor winding circuit having resistance of 30 ohm and inductive reactance of 40 ohm when connected across 230V AC supply.**
Que.10	With the help of diagram show how to get a single-phase supply from a three-phase supply system. Three phase load consist of 100 ohm resistance in each branch and is connected to 3 Ph, 415V, 50Hz supply. Calculate phase voltage, line current, and active power, when load is connected in star and then in delta.
Que.11	Define inductive and capacitive reactance. A voltage of 230 volt is applied across inductance of 350mH and capacitance of 25μF independently. Calculate RMS and maximum value of current flowing through each element

Que.12	Derive the expression for average power in case of R-C series circuit. Draw voltage, current and power waveforms.
Que.13	Derive the relation between line current and phase current for three phase delta connected balanced R-L load.
Que.14	Draw impedance triangle for R-L and R-C series circuit. Resistance of 15Ω inductance of 200mH and capacitance of 150 μF are connected in series across 230V, 50 Hz single phase supply. Calculate impedance, current, power factor and power consumed.
Que.15	Define phase sequence and balanced load. Three phase star connected balanced load with impedance of 9+j12 Ω per phase is connected across 440V, 50 Hz three phase supply. Calculate phase voltage, phase current, line current, active power and reactive power.
Que.16	To the circuit shown in fig. consisting of a 1k ohm resistor connected in series with a 50mH coil, a 10V rms, 10KHz signal is applied. Find: i) impedance Z, ii) current I, iii) Phase angle θ & iv) Voltage across R & L. 
Que.17	**Derive expression for resonance frequency in RLC series circuit and show how reactance's, impedance and current vary with frequency.**
Que.18	State the relation between: i) Phase voltage and line voltage ii) Phase current and line current in case of balanced delta connected 3-ph load. Using above relations, obtain the expressions for 3-ph active power and 3-ph reactive power
Que.19	Three identical impedances each of 6+j8 Ω are connected in star across 3-ph, 415 V, 50 Hz ac supply. Determine i) Line voltage ii) Phase voltage iii) Phase current and line current iv) 3-ph active, reactive and apparent power
Que.20	Explain the concept of lagging taking two electrical quantities with the help of their waveforms and phasor diagrams.
Que.21	The series circuit having resistance 10 Ω and capacitance 150 μF draws a current of 9.4 A from 1-phase, 50 Hz AC supply. Calculate - i) Capacitive reactance ii) impedance iii) power factor iv) supply voltage v) Active power and vi) reactive power.
Que.22	Derive the relation between i) phase voltage and line voltage ii) phase current and line current in case of balanced STAR connected 3-ph inductive load. Assume phase sequence RYB. Draw the circuit diagram & necessary phasor diagram.
Que.23	A series R-L-C circuit consists of R = 10 Ω, L = 0.318 H and C = 63.6μF. This circuit is supplied by source of emf given by $e(t)$ = 100 sin314t. Find: i) Expression for i (t) ii) Phase angle between voltage and current iii) Power factor of circuit iv) Active power consumed

Que.24	A sinusoidally varying alternating voltage of 100 V (rms value) with 50 Hz frequency is applied to a circuit find : i) The mathematical equation of the voltage; ii) Time Period iii) The instantaneous voltage when t = 1.667 ms; iv) The time when instantaneous voltage is 100 V; v) Average value of the voltage vi) Maximum value of the voltage.
Que.25	Three sinusoidally alternating currents varying at 50Hz have rms values 5 A, 7.5 A and 10 A with phase angles of 30º, – 60º and 45º respectively. i) Find their average values ii) Write equations for their instantaneous values iii) Find the instantaneous values of 10 A current at 100 mSec.